

GET 324 Mini-Project Report

Cloud Computing and AI Model Deployment for Engineering Applications
Group ME9 | Mechanical Engineering

Assigned Task

Binary image classification: **Tomato Leaf Mold** vs **Tomato Septoria Leaf Spot**

Project Report

The dataset for this binary classification task, Tomato Leaf Mold versus Tomato Septoria Leaf Spot, was sourced from the PlantVillage dataset on Kaggle, comprising 2,179 training images and 544 validation images across the two classes. A MobileNetV2 model, pre-trained on ImageNet, was fine-tuned on these images and achieved a final validation accuracy of 96.88 percent. The trained model was deployed as an interactive Streamlit web application, allowing users to upload a tomato leaf photo and instantly receive a predicted diagnosis with a confidence score. During development, the team encountered a corrupted cached weights file, a missing Streamlit dependency, and a file-path mismatch that caused the deployed app to fail loading the model. These were resolved by clearing the cache, installing missing packages, and ensuring consistent working directories. Future improvements could include expanding to more disease classes and adding explainability visualizations.

Model Performance Summary

Metric	Value
Base Architecture	MobileNetV2 (ImageNet pre-trained)
Training Images	2,179
Validation Images	544
Final Validation Accuracy	96.88%
Final Validation Loss	0.1144

Project Links

GitHub Repository: <https://github.com/robinson-wisdom172006/ME9>

Deployed Application: <https://groupme9.streamlit.app/>

Team Members

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